

Patterns

1) for the outer loop, count the number of lines.

2) for inner loop, focus on the columns, and connect them somehow to the rows.

3) print them inside the inner for loop

4) observe symmetry. (optional)

1) * * * * *

* * * * *

* * * * *

* * * * *

* * * * *

simple pattern

```
code:  for(i=0;i<5;i++)
        {
            for(j=0;j<5;j++)
            {
                print(*)
            }
        }
```

2) left triangle pattern

*

* *

* * *

* * * *

* * * * *

```
code:  for (int i = 0; i < n; i++)
        {
            for (int j = 0; j <= i; j++)
            {
                cout << " * ";
            }
            cout << endl;
        }
```

3) left triangle with numbers

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

```
code:  for (int i = 1; i <= n; i++)
        {
            for (int j = 1; j <= i; j++)
            {
                cout << " " << j << " ";
            }
            cout << endl;
        }
```

if inner loop print i then we get row number at every row elements.

4) downward triangle pattern:

code:

```
* * * * *
* * * *
* * *
* *
*
```

```
for (int i = 1; i <= n; i++)
{
    for (int j = 1; j <= n - i + 1; j++)
    {
        cout << " * ";
    }
    cout << endl;
}
```

5) upward pyramid

for 0 based indexing

```
 *
***
*****
*****
*****
```

[4 space 1 star 4 space]
[3 space 3 star 3 space]
[2 space 5 star 2 space]
[1 space 7star 1 space]
[0 space 9 star 0 space]

ie. in general $n-i-1$ spaces left and right

$2*i + 1$ stars

```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j <= n - i - 1; j++)
    {
        cout << " ";
    }
    for (int j = 0; j < 2 * i + 1; j++)
    {
        cout << "*";
    }
    for (int j = 0; j <= n - i - 1; j++)
    {
        cout << " ";
    }
    cout << endl;
}
```

6) down pyramid

code: for (int i = 0; i < n; i++)

```
*****
*****
*****
***
*
```

```
{
    for (int j = 0; j < i; j++)
    {
        cout << " ";
    }
    for (int j = 0; j < 2 * n - (2 * i + 1); j++)
    {
        cout << "*";
    }
    for (int j = 0; j < i; j++)
    {
        cout << " ";
    }
    cout << endl;
}
```

7) vertical triangle

code: for (int i = 1; i <= 2 * n - 1; i++)

```
{
    int stars = i;
    if (i > n)
    {
        stars = 2 * n - i;
    }
    for (int j = 1; j <= stars; j++)
    {
        cout << "*";
    }
    cout << endl;
}
```

8) staircase pattern

code: int space = 2 * (n - 1);

```
for (int i = 1; i <= n; i++)
{
    for (int j = 1; j <= i; j++)
    {
        cout << j;
    }
    for (int j = 1; j <= space; j++)
    {
        cout << " ";
    }
    for (int j = i; j >= 1; j--)
    {
        cout << j;
    }
    cout << endl;
    space -= 2;
}
```

1 1
12 21
123 321
1234 4321
1234554321

spaces are $2n-2$, also in each next row

we have to reduce spaces by 2

9) char pattern

```
A
ABA
ABCBA
ABCD CBA
ABCDEDCBA
```

```
code;
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n - i - 1; j++)
    {
        cout << " ";
    }
    char ch = 'A';
    int breakpoint = (2 * i + 1) / 2;
    for (int j = 0; j < 2 * i + 1; j++)
    {
        cout << ch;
        if (j < breakpoint)
            ch++;
        else
            ch--;
    }
    for (int j = 0; j < n - i - 1; j++)
    {
        cout << " ";
    }
    cout << endl;
}
```

10)

```
*****
****  ****
***   ***
**    **
*     *
*     *
**    **
***   ***
****  ****
*****
*****
```

code:

```

for (int i = 0; i < n; i++)
{
    for (int j = 1; j <= n - i; j++)
    {
        cout << "*";
    }
    for (int j = 0; j < 2 * i; j++)
    {
        cout << " ";
    }
    for (int j = 1; j <= n - i; j++)
    {
        cout << "*";
    }
    cout << endl;
}
for (int i = 1; i <= n; i++)
{
    for (int j = 1; j <= i; j++)
    {
        cout << "*";
    }
    for (int j = 0; j < (2 * n) - 2 * i; j++)
    {
        cout << " ";
    }
    for (int j = 1; j <= i; j++)
    {
        cout << "*";
    }
    cout << endl;
}

```

11) butterfly pattern

```

*          *
**         **
***        ***
****       ****
*****      *****
*****      *****
****       ****
***        ***
**         **
*          *

```

```

void butterflypattern(int n)
{
    int spaces = 2 * n - 2;
    for (int i = 1; i <= 2 * n - 1; i++)
    {
        int stars = i;
        if (i > n)
            stars = 2 * n - i;
        for (int j = 1; j <= stars; j++)
        {
            cout << "*";
        }
        for (int j = 1; j <= spaces; j++)
        {
            cout << " ";
        }
        for (int j = 1; j <= stars; j++)
        {
            cout << "*";
        }
        cout << endl;
        if (i < n)
        {
            spaces -= 2;
        }
        else
            spaces += 2;
    }
}

```

12)

```

4 4 4 4 4 4 4
4 3 3 3 3 3 4
4 3 2 2 2 3 4
4 3 2 1 2 3 4
4 3 2 2 2 3 4
4 3 3 3 3 3 4
4 4 4 4 4 4 4

```

```

for (int i = 0; i < 2 * n - 1; i++)
{
    for (int j = 0; j < 2 * n - 1; j++)
    {
        int top = i;
        int left = j;
        int right = (2 * n - 2) - j;
        int down = (2 * n - 2) - i;
        cout << (n - min(min(top, down), min(left, right))) << " ";
    }
    cout << endl;
}

```